

File permissions in Linux

Project description

This project demonstrates the implementation of access control in a Linux environment through the management of file and directory permissions. Using command-line tools such as `chmod` and `ls -la`, permissions were audited and adjusted to ensure that only authorized research team members could access specific resources. The project applies the principle of least privilege to enhance data security, prevent unauthorized access, and support secure system operations.

Check file and directory details

After using 'cd projects' to navigate to the target folder, the 'ls' command listed all the files in the folder. The 'ls -la' command was used inside the projects folder and returned all files and file permissions in the directory, including the hidden file "project_x.txt"

```
researcher2@384ca379e5b2:~$ pwd
/home/researcher2
researcher2@384ca379e5b2:~$ ls
projects
researcher2@384ca379e5b2:~$ cd projects
researcher2@384ca379e5b2:~/projects$ ls
drafts          project_m.txt  project_t.txt
project_k.txt   project_r.txt
researcher2@384ca379e5b2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May  4 11:22 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 11:36 ..
-rw--w---- 1 researcher2 research_team   46 May  4 11:22 .project_
x.txt
drwx--x--- 2 researcher2 research_team 4096 May  4 11:22 drafts
-rw-rw-rw- 1 researcher2 research_team   46 May  4 11:22 project_k
.txt
-rw-r----- 1 researcher2 research_team   46 May  4 11:22 project_m
.txt
-rw-rw-r-- 1 researcher2 research_team   46 May  4 11:22 project_r
.txt
-rw-rw-r-- 1 researcher2 research_team   46 May  4 11:22 project_t
.txt
researcher2@384ca379e5b2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 May  4 11:22 drafts
```

Describe the permissions string

The permission string in file authorization is a 10-character string which determines if a file has read, write and execute permissions. 10-character string begins each entry and indicates how the permissions on the file are set. For instance, a directory with full permissions for all owner types would be `drwxrwxrwx`. (d) stands for directory, (r) stands for read, (w) stands for write, (x) stands for execute. Each of the three segments of permissions represent the user, group and other.

```
researcher2@384ca379e5b2:~$ pwd
/home/researcher2
researcher2@384ca379e5b2:~$ ls
projects
researcher2@384ca379e5b2:~$ cd projects
researcher2@384ca379e5b2:~/projects$ ls
drafts          project_m.txt  project_t.txt
project_k.txt  project_r.txt
researcher2@384ca379e5b2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May  4 11:22 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 11:36 ..
-rw--w---- 1 researcher2 research_team   46 May  4 11:22 .project_
x.txt
drwx--x--- 2 researcher2 research_team 4096 May  4 11:22 drafts
-rw-rw-rw- 1 researcher2 research_team   46 May  4 11:22 project_k
.txt
-rw-r----- 1 researcher2 research_team   46 May  4 11:22 project_m
.txt
-rw-rw-r-- 1 researcher2 research_team   46 May  4 11:22 project_r
.txt
-rw-rw-r-- 1 researcher2 research_team   46 May  4 11:22 project_t
.txt
researcher2@384ca379e5b2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 May  4 11:22 drafts
```

Change file permissions

A file “project_m.txt” was determined to have group permissions which were not authorized and therefore had to be changed. The command ‘`chmod g-r project_m.txt`’ was used to remove the group read access. This change will prevent all other segments or entries except the user from reading or writing the file.

```
researcher2@384ca379e5b2:~/projects$ chmod g-r project_m.txt
researcher2@384ca379e5b2:~/projects$ ls -la project_m.txt
-rw----- 1 researcher2 research_team 46 May  4 11:22 project_m.t
xt
researcher2@384ca379e5b2:~/projects$
```

Change file permissions on a hidden file

The file `.project_x.txt` is an archived file and thus hidden. By default it should not have write access by any segment but the user and group can read the file. To fix this, the command `'chmod u-w,g-w,g+r .project_x.txt'` is used. This removes write permissions for both user and group segments but grants read only access. The `(.)` in front of the file name indicates that it is a hidden file.

```
researcher2@d2880d9c3fa4:~/projects$ chmod u-w,g-w,g+r .project_x.
txt
researcher2@d2880d9c3fa4:~/projects$ ls -la .project_x.txt
-r--r----- 1 researcher2 research_team 46 May  4 11:36 .project_x.
txt
researcher2@d2880d9c3fa4:~/projects$
```

Change directory permissions

A directory `'drafts'` was determined to be accessible by a user other than the primary user "researcher2", to fix this the command `'chmod g-x drafts'` was used to remove the group execute permission.

```
researcher2@d2880d9c3fa4:~/projects$ ls
drafts      project_m.txt  project_t.txt
project_k.txt  project_r.txt
researcher2@d2880d9c3fa4:~/projects$ ls -la drafts
total 8
drwx--x--- 2 researcher2 research_team 4096 May  4 11:36 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 11:36 ..
researcher2@d2880d9c3fa4:~/projects$ chmod g-x drafts
researcher2@d2880d9c3fa4:~/projects$ ls -la drafts
total 8
drwx----- 2 researcher2 research_team 4096 May  4 11:36 .
drwxr-xr-x 3 researcher2 research_team 4096 May  4 11:36 ..
researcher2@d2880d9c3fa4:~/projects$
```

Summary

This project demonstrates the use of Linux command-line tools to manage user authorization and file permissions within a system environment. The objective was to enforce proper access control by ensuring that members of a research team were granted only the permissions necessary for their roles.

Using key commands such as `chmod` and `ls -la`, file and directory permissions were analyzed, modified, and validated to align with the principle of least privilege. The project highlights practical implementation of access control mechanisms, strengthening system security and reducing the risk of unauthorized access or data exposure.

Overall, this work reflects foundational skills in system administration and security, with a focus on maintaining data integrity and supporting secure organizational operations.
